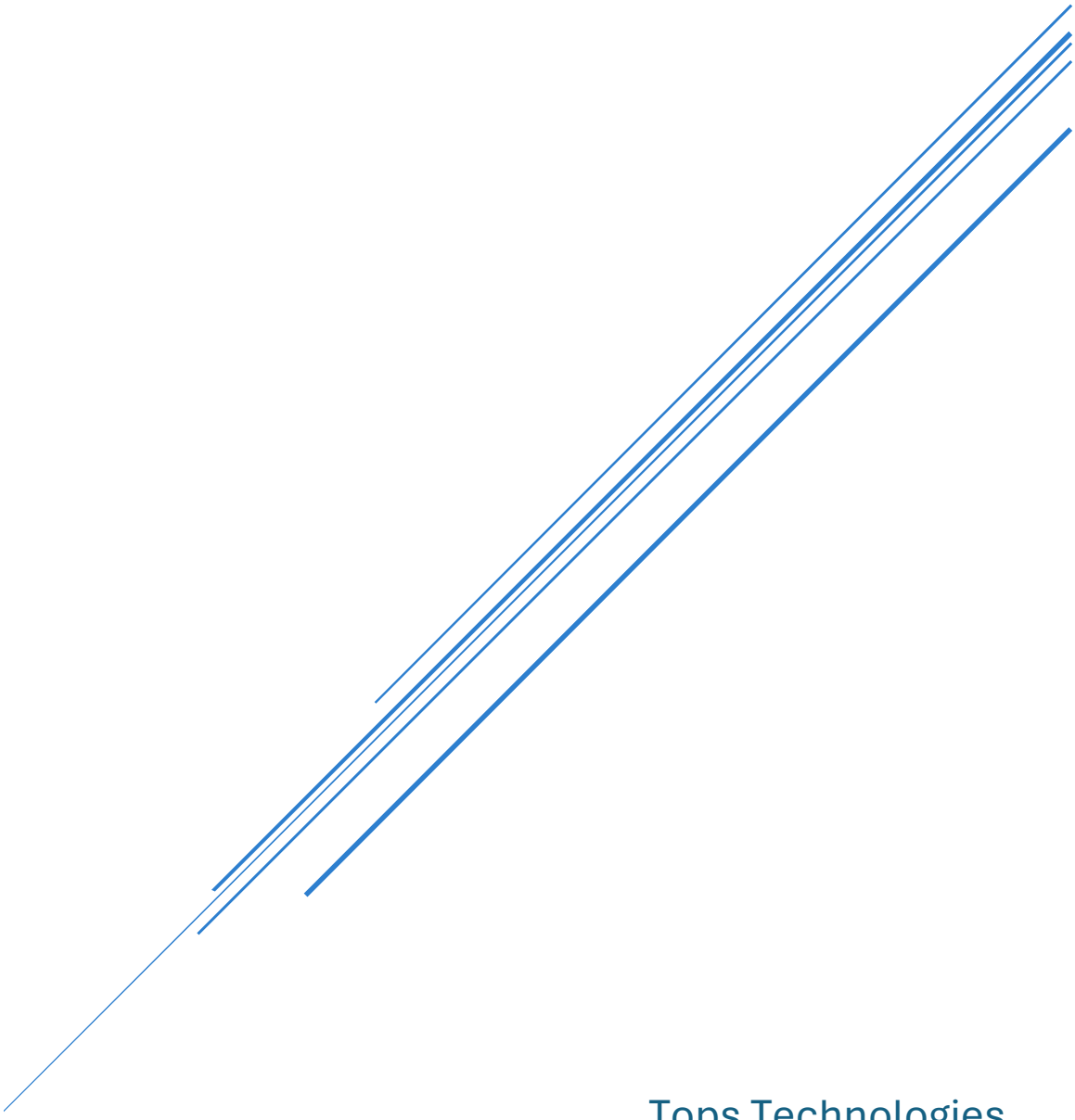
DAY – 5 NETWORK TESTING TOOLS

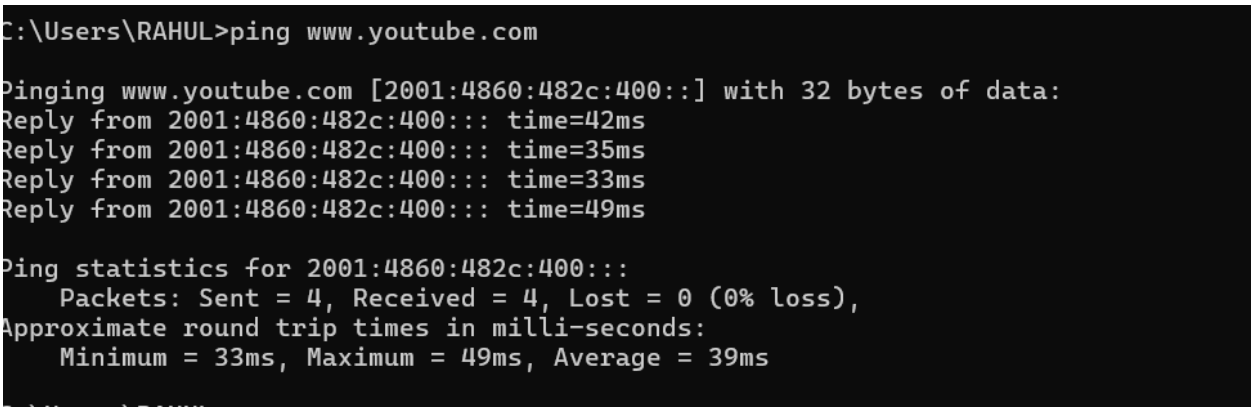
Assignment - 16



Tops Technologies
Cyber Security

1. **Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.**

➢



2. **Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.**
   Hint: Each hop represents a network device your data passes through to reach the destination.

➢

```
C:\Users\RAHUL>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.63.84.208]
over a maximum of 30 hops:

  1     5 ms     1 ms     1 ms  Unit [192.168.1.1]
  2    33 ms    33 ms    20 ms  10.240.15.231
  3     *        *        *     Request timed out.
  4    29 ms    28 ms    38 ms  dsl-ncr-dynamic-101.89.23.125.airtelbroadband.in [125.23.89.101]
  5    50 ms    90 ms   113 ms  116.119.46.40
  6    67 ms   101 ms    95 ms  ae9.tata-hyd.netarch.akamai.com [23.65.124.252]
  7    62 ms    54 ms    65 ms  a23-63-84-208.deploy.static.akamaitechnologies.com [23.63.84.208]

Trace complete.
```

3. **Use the nslookup tool to find the IP address of www.zomato.com and record the result.**

➢



4. **Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.**

➢

```
C:\Users\RAHUL>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=87ms TTL=64
Reply from 192.168.1.2: bytes=32 time=6ms TTL=64
Reply from 192.168.1.2: bytes=32 time=120ms TTL=64
Reply from 192.168.1.2: bytes=32 time=68ms TTL=64

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 120ms, Average = 70ms
```

5. **Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.**
   Constraint: Give a specific example for each tool based on your own test results above.

➢

Network Troubleshooting Report

1. **Website not loading: The ping tool can check connectivity to a website's server, for example, ping www.zomato.com. If it fails, the server may be unreachable or blocking ping requests.**
2. **Slow connection: The traceroute tool helps identify where delays occur along the network path. For example, tracert www.zomato.com shows the route and response times at each hop.**
3. **Suspecting DNS issues: The nslookup tool checks whether a domain name resolves to an IP address. For example, nslookup www.zomato.com displays the DNS server and the resolved IP address.**
4. **In my DNS test, I used nslookup www.zomato.com to check the domain's IP address.**
5. **In my connectivity test, I used ping with another device's IP address to check whether it responded.**
6. **These tools help diagnose different network problems, but their results must be interpreted carefully because firewalls and server configurations can affect responses.**